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Exp No: 1	Study of various Network commands used

AIM:

To study and practise the various network commands used in Linux and Windows operating systems.

ALGORITHM / PROCEDURE:

1. Open the Terminal in Linux and the Command Prompt (`cmd`) in Windows.
2. Use `hostname` to see the name of the computer.
3. Use `ifconfig` (Linux) and `ipconfig` (Windows) to see the IP address, subnet mask and MAC address.
4. Use `ping` to check whether the remote system is reachable. Note the packet loss and the time taken.
5. Use `traceroute` (Linux) and `tracert` (Windows) to see the path taken by the packets.
6. Use `nslookup` and `dig` to find the IP address of a domain name.
7. Use `netstat` to see the active connections and open ports.
8. Use `arp -a` to see the IP address to MAC address table.
9. Use `route` (Linux) and `route print` (Windows) to see the routing table.
10. Note down the output of each command.

Commands used:

Purpose	Linux	Windows
Host name	`hostname`	`hostname`
IP configuration	`ifconfig / ip addr`	`ipconfig /all`
Check connectivity	`ping -c 4 google.com`	`ping google.com`
Trace the route	`traceroute google.com`	`tracert google.com`
DNS lookup	`nslookup / dig`	`nslookup`
Active connections	`netstat -tuln`	`netstat -an`
ARP table	`arp -a`	`arp -a`
Routing table	`route -n`	`route print`
MAC address	`ip link`	`getmac`

DIAGRAM:

Fig 1: Network commands and the layers they work in

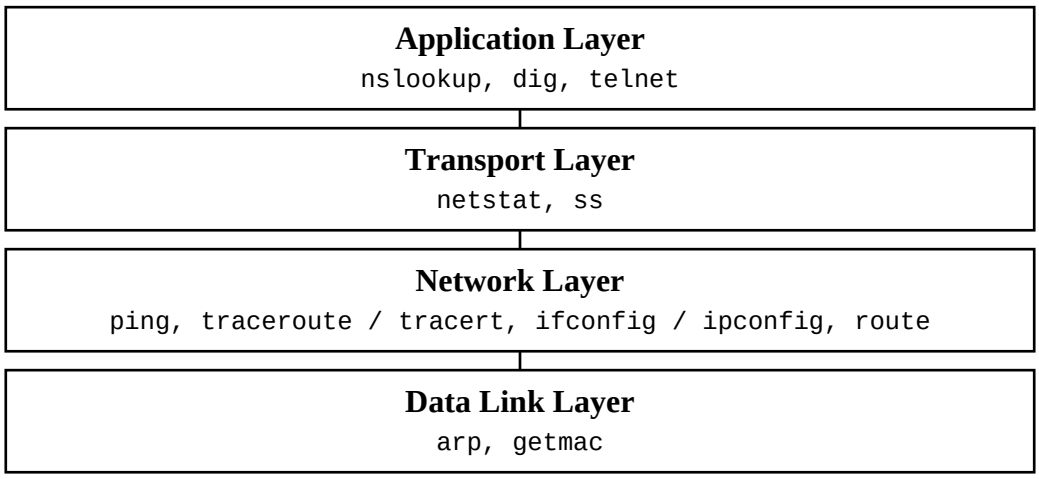
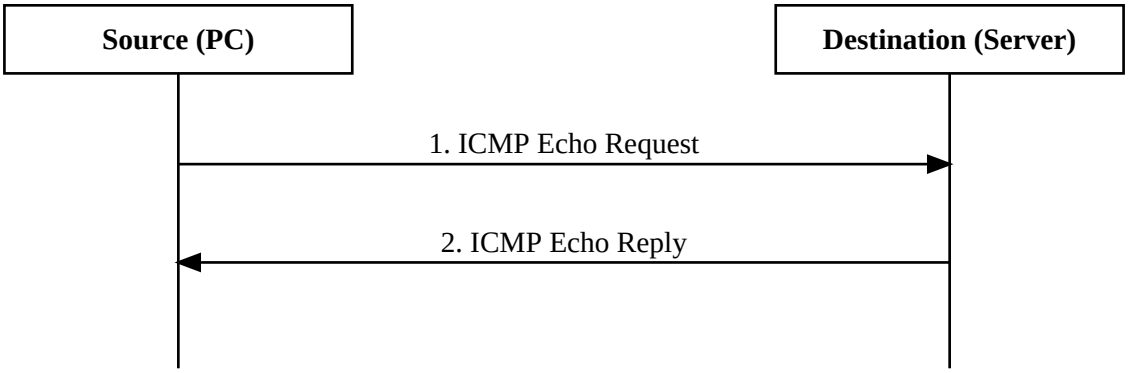


Fig 2: Working of ping command



OUTPUT:

Linux:

```
USER @USER -VirtualBox: ~
```

```
USER   @USER   -VirtualBox:~$ hostname
USER   -VirtualBox
```

```
USER @USER -VirtualBox: ~
```

```
USER   @USER   -VirtualBox:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.1.10  netmask 255.255.255.0  broadcast 192.168.1.255
        inet6 fe80::a00:27ff:fe4e:66a1  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:4e:66:a1  txqueuelen 1000  (Ethernet)
        RX packets 2543  bytes 2154321 (2.1 MB)
        TX packets 1987  bytes 312456 (312.4 KB)

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
```

USER @USER -VirtualBox: ~

```
USER   @USER   -VirtualBox:~$ ping -c 4 google.com
PING google.com (142.250.195.78) 56(84) bytes of data.
64 bytes from maa03s45-in-f14.1e100.net (142.250.195.78): icmp_seq=1 ttl=117 time=14.2 ms
64 bytes from maa03s45-in-f14.1e100.net (142.250.195.78): icmp_seq=2 ttl=117 time=14.8 ms
64 bytes from maa03s45-in-f14.1e100.net (142.250.195.78): icmp_seq=3 ttl=117 time=13.9 ms
64 bytes from maa03s45-in-f14.1e100.net (142.250.195.78): icmp_seq=4 ttl=117 time=15.2 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 13.904/14.525/15.230/0.512 ms
```

USER @USER -VirtualBox: ~

```
USER   @USER   -VirtualBox:~$ traceroute google.com
traceroute to google.com (142.250.195.78), 30 hops max, 60 byte packets
 1  _gateway (192.168.1.1)  1.102 ms  1.045 ms  0.987 ms
 2  10.20.0.1 (10.20.0.1)  4.321 ms  4.287 ms  4.410 ms
 3  172.16.5.9 (172.16.5.9)  8.765 ms  8.801 ms  8.730 ms
 4  72.14.215.85 (72.14.215.85)  12.543 ms  12.498 ms  12.610 ms
 5  maa03s45-in-f14.1e100.net (142.250.195.78)  14.230 ms  14.187 ms  14.301 ms
```

USER @USER -VirtualBox: ~

```
USER   @USER   -VirtualBox:~$ nslookup google.com
Server:         192.168.1.1
Address:        192.168.1.1#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.195.78
```

USER @USER -VirtualBox: ~

```
USER   @USER   -VirtualBox:~$ dig google.com +short
142.250.195.78
```

USER @USER -VirtualBox: ~

```
USER   @USER   -VirtualBox:~$ netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN
tcp6       0      0 :::80                   :::*                    LISTEN
udp        0      0 0.0.0.0:68              0.0.0.0:*
```

USER @USER -VirtualBox: ~

```
USER   @USER   -VirtualBox:~$ arp -a
_gateway (192.168.1.1) at c4:e9:84:2a:5b:10 [ether] on enp0s3
? (192.168.1.12) at 3c:52:82:7d:11:9e [ether] on enp0s3
```

USER @USER -VirtualBox: ~

```
USER   @USER   -VirtualBox:~$ route -n
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0         192.168.1.1     0.0.0.0         UG    100    0        0 enp0s3
192.168.1.0     0.0.0.0         255.255.255.0   U     100    0        0 enp0s3
```

Windows:

Command Prompt

```
C:\Users\USER   >hostname
DESKTOP-USER
```

Command Prompt

```
C:\Users\USER   >ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : DESKTOP-USER
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No

Ethernet adapter Ethernet:

   Physical Address. . . . . . . . . : 00-1A-2B-3C-4D-5E
   DHCP Enabled. . . . . . . . . . . : Yes
   IPv4 Address. . . . . . . . . . . : 192.168.1.25(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.1.1
   DHCP Server . . . . . . . . . . . : 192.168.1.1
   DNS Servers . . . . . . . . . . . : 8.8.8.8
```

Command Prompt

```
C:\Users\USER   >ping google.com

Pinging google.com [142.250.195.78] with 32 bytes of data:
Reply from 142.250.195.78: bytes=32 time=15ms TTL=117
Reply from 142.250.195.78: bytes=32 time=14ms TTL=117
Reply from 142.250.195.78: bytes=32 time=14ms TTL=117
Reply from 142.250.195.78: bytes=32 time=16ms TTL=117

Ping statistics for 142.250.195.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 14ms, Maximum = 16ms, Average = 14ms
```

Command Prompt

```
C:\Users\USER   >tracert google.com

Tracing route to google.com [142.250.195.78]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  192.168.1.1
  2     4 ms     4 ms     4 ms  10.20.0.1
  3     9 ms     8 ms     9 ms  172.16.5.9
  4    13 ms    12 ms    12 ms  72.14.215.85
  5    14 ms    14 ms    15 ms  142.250.195.78

Trace complete.
```

Command Prompt

```
C:\Users\USER   >nslookup google.com
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:    google.com
Address:  142.250.195.78
```

Command Prompt

```
C:\Users\USER   >netstat -an

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:445            0.0.0.0:0              LISTENING
  TCP    192.168.1.25:50234     142.250.195.78:443     ESTABLISHED
  UDP    0.0.0.0:5353           *:*
```

Command Prompt

```
C:\Users\USER   >arp -a

Interface: 192.168.1.25 --- 0x8
  Internet Address      Physical Address      Type
  192.168.1.1           c4-e9-84-2a-5b-10     dynamic
  192.168.1.255         ff-ff-ff-ff-ff-ff     static
```

Command Prompt

```
C:\Users\USER   >route print

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      192.168.1.1    192.168.1.25     25
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
      192.168.1.0    255.255.255.0         On-link      192.168.1.25    281
```

Command Prompt

```
C:\Users\USER   >getmac

Physical Address    Transport Name
=================== ==========================================================
00-1A-2B-3C-4D-5E   \Device\Tcpip_{8F3A21B0-5C6D-4E7F-9A1B-2C3D4E5F6A7B}
```

RESULT:

Thus the various network commands used in Linux and Windows were studied and executed successfully.